

SECUREBALLOT: RFID-BASED SECURE ELECTRONIC VOTING SYSTEM

AIM:

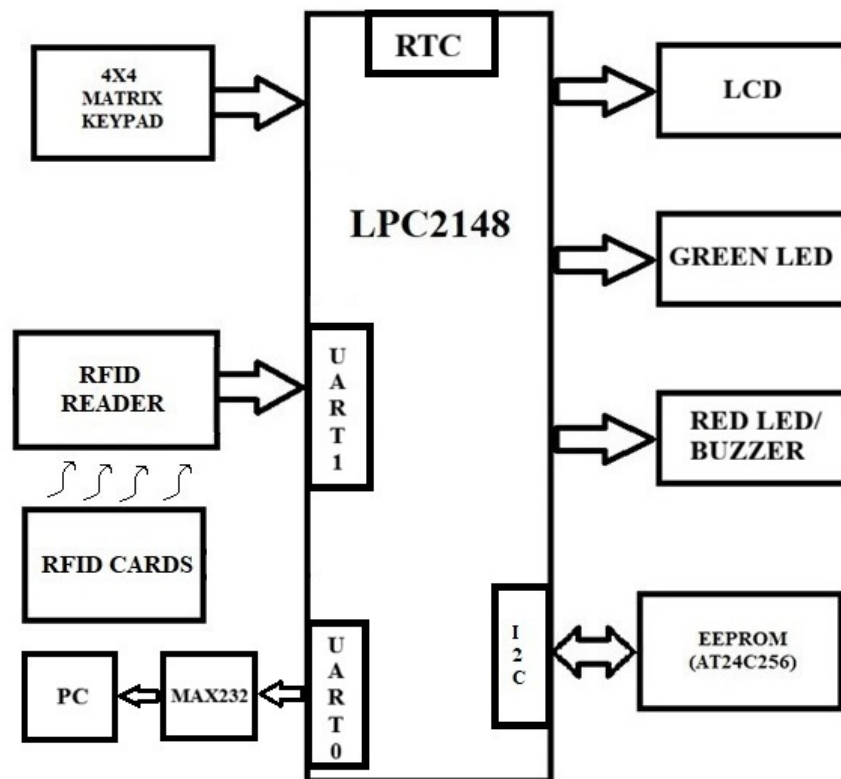
The aim of SecureBallot: RFID-Based Secure Electronic Voting System is to design and develop a secure, reliable, and user-friendly electronic voting system that utilizes RFID technology for voter authentication and embedded systems for vote management. The system aims to ensure that only authorized voters can cast their votes, prevent duplicate voting through a one-voter-one-vote mechanism, and provide accurate, transparent, and tamper-resistant election results.

The project also seeks to automate the complete voting process, including voter verification, candidate selection, vote recording, result generation, and election administration, thereby reducing manual effort, minimizing human errors, enhancing election security, and improving the overall efficiency and integrity of the voting process.

INSIGHT:

- Good knowledge of C-Programming.
- Knowledge of Embedded-C programming
- Thorough with the LPC2148 Architecture, General purpose I/O Interface
- Understanding of RFID usage.
- Understanding of I2C protocol.
- Understanding of LCD and KEYPAD.

BLOCK DIAGRAM:



REQUIREMENTS:

HARDWARE REQUIREMENTS:

- LPC 2148
- RFID READER
- RFID CARDS
- 20X4 LCD
- 4X4 MATRIX KEYPAD
- BUZZER OR LEDS
- AT24C256
- MAX232
- USB TO UART CONVERTER

SOFTWARE REQUIREMENTS:

- EMBEDDED C – PROGRAMMING
- KEIL-C COMPILER
- FLASH MAGIC

SEQUENCE TO BE FOLLOWED FOR IMPLEMENTATION:

- Create a new folder in your Server / Laptop / PC and save that folder with your project name.
- Copy all developed driver files into the project folder such as lcd.c, lcd.h, delay.c, delay.h, keypad.c, keypad.h, uart0.c, uart0.h, uart1.c, uart1.h, i2c.c, i2c.h, eeprom.c, eeprom.h, rtc.c and rtc.h.
- Individually verify each module before integrating the complete system.
- First check LCD module by displaying character constant, string constant and integer constant.
- Next check keypad peripheral by displaying the pressed key values on LCD.
- Next check UART peripheral to transmit character constant, string constant and integer constant and to receive string constant using UART interrupts.
- Connect RFID reader D0 pin to USB-to-UART converter and test the RFID reader working condition at 9600 baud rate OR select the required hardware connection to test the RFID reader is working or not.
- Write logic to read RFID card number using UART1 interrupt and display the card number on LCD.
- Test I2C communication and verify EEPROM read and write operations.
- Test on-chip RTC module, set date and time once and verify by displaying real-time clock values on LCD.
- After verifying all individual modules, integrate LCD, keypad, RFID, UART0, UART1, EEPROM, RTC and serial logging into a single application.
- On power ON, initialize LCD, keypad, UART0, UART1, EEPROM, I2C, RTC

and serial logger modules.

- Display the project name on LCD for a few seconds.
- Display current date and time from RTC on LCD for once to test RTC info is correct or not.
- Display “Waiting for Card” message on the LCD.
- When an RFID card is placed in front of the reader, the reader transmits the card number through serial communication at 9600 baud rate.
- Controller reads the RFID card number using UART interrupt.
- Controller reads voting start time, voting end time and voting enable flag from EEPROM.
- Controller reads current time from RTC and compares it with stored voting time duration.
- If voting status is OFF or current time is outside the allowed voting duration, display “Voting Closed – Access Denied” on LCD, turn ON red LED and return to waiting state.
- If voting status is ON and current time is within the valid voting duration, proceed for voter authentication.
- If the RFID card matches predefined voter IDs stored in EEPROM, display voter identification message such as “Voter1” on LCD.

Display voter menu on LCD:

1. VOTING
2. EDIT PASSWORD
3. EXIT

- If option 1 is selected, prompt the voter to enter the password using keypad.
- For every key press, display ‘*’ instead of actual number to hide the password.
- After password entry, voter presses enter key (#).
- If password is incorrect, turn ON red LED, display wrong password message and continue for next voter.
- If password is correct, display party selection menu with party symbols created

using LCD CGRAM.

- Based on voter selection, read the corresponding party vote count from EEPROM, increment it by one and store the updated value back into EEPROM.
- Display “Vote Casted Successfully” message and turn ON green LED for confirmation.
- If option 2 is selected, prompt the voter to enter old password, new password and confirm password.
- After successful password validation, store updated password into EEPROM and display confirmation message.
- If option 3 is selected, exit voter menu and return to waiting for card state.
- Repeat the above voter process continuously for different voters during the valid voting time duration.
- Assign a separate RFID card for the election officer. If election officer card will accept only external interrupt is requested. Based on external interrupt request asking for officer card. In this place other cards are neglected.

When officer card is detected, display officer menu on LCD:

1. SET VOTING TIME
2. START VOTING
3. STOP VOTING
4. VIEW RESULT
5. RESET VOTING
6. RTC EDIT
7. EXIT

- If officer selects set voting time option, prompt to enter voting start hour, start minute, end hour and end minute using keypad.
- Validate the entered timing values and store them into EEPROM.
- If officer selects start voting option, ask for officer security password and set voting enable flag in EEPROM.
- If officer selects stop voting option, ask for officer password and clear voting

enable flag in EEPROM.

- If officer selects view result option, ask for officer password, read all party vote counts from EEPROM and display the results on LCD.
- If officer selects reset voting option, ask for officer password, reset all party vote counts and voting status flag in EEPROM.
- If officer is selected the RTC edit, then officer is allowed to set the time and date through keypad to maintain the exact time.
- If officer selects exit option, return to waiting for card state.
- RTC and EEPROM ensure that voting time and voting status are retained even after power failure.
- Voting outside the authorized time duration is completely blocked to prevent illegal voting.
- Configure UART1 at 9600 baud to receive the 10-byte RFID card frame from the RFID reader using interrupts.
- Configure UART0 as a separate communication channel to transmit project status and audit information to the PC serial terminal.
- Connect UART0 of LPC2148 to the PC through MAX232 and a USB-to-UART converter, as shown in the block diagram.
- Open the serial terminal on the PC, select the detected COM port, select the configured baud rate and establish the connection.
- Create a serial logger module to transmit every event in the format Date, Time, Event, Card ID, Status and Description.
- Use the on-chip RTC to attach the current date and time to every transmitted log entry.
- After system initialization, transmit a SYSTEM START log message to confirm that the controller and serial terminal are operating correctly.
- Whenever an RFID card is detected, transmit a CARD SCAN log containing the card number or a masked card number and the authentication status.
- If voting is disabled or the current RTC time is outside the authorized voting

duration, transmit a VOTING ACCESS BLOCKED log before returning to the waiting state.

- If an unregistered RFID card is detected, transmit an AUTHENTICATION FAILED log and activate the red LED or buzzer.
- If an already-used voter card is presented, reject the request and transmit a DUPLICATE VOTING ATTEMPT log.
- If the voter enters an incorrect password, transmit a PASSWORD CHECK FAILED log without recording the entered password.
- After a vote is successfully stored in EEPROM, transmit only a VOTE CAST SUCCESS log. Do not transmit or store the selected candidate or party along with the voter card number, to preserve ballot secrecy.
- Transmit separate audit logs when the election officer sets voting time, starts voting, stops voting, views results, resets voting data, edits RTC information or exits the officer menu.
- Develop the PC serial-terminal application to continuously receive complete UART lines, display them in real time and append them to a CSV or text log file.
- Create a separate date-and-time-stamped log file for every election session, for example SecureBallot_Log_2026-07-14_09-00-00.csv.
- Provide terminal controls for COM-port selection, baud-rate selection, connect, disconnect, clear terminal, start logging, stop logging and open log file.
- Test serial logging for system start, valid voter, invalid card, wrong password, duplicate voting attempt, voting outside the permitted time and all officer operations.
- Verify that the EEPROM retains vote counts, voter status and voting configuration, while the PC log file contains only system, authentication, security and administrative audit events.
- Close the serial log file properly when voting is stopped or the terminal application is disconnected, so that all received information is safely stored.

➤ If you are getting this output, then your project is completed.

***** ALL THE BEST *****

Log File info:

SecureBallot - RFID Based Secure Electronic Voting System Election Audit Log

Election ID : EC2026-001
Polling Booth : Booth-08
Polling Officer : 99990001
Date : DD-MM-YYYY

DATE, TIME, EVENT, CARD_ID, STATUS, DESCRIPTION

14-07-2026, 08:55:10, SYSTEM_START, -, SUCCESS, System initialized
14-07-2026, 08:56:10, OFFICER_LOGIN,99990001, SUCCESS, Officer authenticated
14-07-2026, 08:56:45, START_VOTING, 99990001, SUCCESS, Voting enabled

14-07-2026, 09:01:12, CARD_SCAN, 12345678, RECEIVED, RFID detected
14-07-2026, 09:01:14, AUTHENTICATION,12345678, SUCCESS, Voter authenticated
14-07-2026, 09:01:22, PASSWORD_CHECK,12345678, SUCCESS, Password verified
14-07-2026, 09:01:36, VOTE_CAST,12345678, SUCCESS, Vote recorded

14-07-2026, 09:12:45, CARD_SCAN, 87651234, RECEIVED, RFID detected
14-07-2026, 09:12:47, AUTHENTICATION, 87651234, FAILED, Invalid card

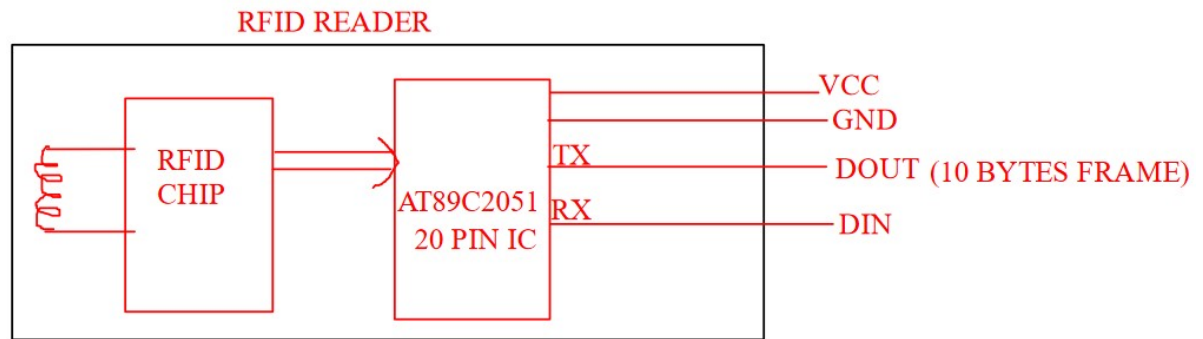
14-07-2026, 09:35:16, CARD_SCAN, 12345678, RECEIVED, RFID detected
14-07-2026, 09:35:18, DUPLICATE_VOTE, 12345678, BLOCKED, already voted

14-07-2026, 17:30:10, VIEW_RESULT, 99990001, SUCCESS, Results viewed
14-07-2026, 17:31:00, STOP_VOTING, 99990001, SUCCESS, Voting stopped

End of Election Audit Log

Note: To make the application more effective and realistic, please review it thoroughly and incorporate all necessary enhancements, improvements, and refinements wherever required.

RFID READER: RFID Reader will send 10 bytes of data once card is placed nearer to the RFID Reader. Once check the below images to get some idea on RFID Reader block diagram. Refer the supported data sheets to get some more knowledge.



For example, card number is 12345678, then output of the RFID Reader is
0x02 0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x03 (hex format)